CHAPTER 1

Introduction to Cognitive Science

Learning Objectives

By the end of this chapter, you should be able to:

(1) explain the difference between *cognition* and *cognitive science*;

(2) recognise key cognitive processes in a familiar activity;

(3) name the disciplines that contribute to cognitive science;

(4) compare symbolic, connectionist, and probabilistic models using a simple example;

(5) explain why cognitive science is relevant to artificial intelligence (AI).

1.1 Why study cognition?

Imagine arriving at university and seeing a message: "Today's AI lecture is in Room 204." You read the message, remember where the building is, follow signs, and decide where to turn. If you reach the wrong room, you check the information and change your route. This ordinary task uses several mental processes working together.

Key Idea

The central question of this chapter: How do people obtain information, understand it, remember it, and use it to choose an action? Cognitive science investigates these abilities; AI develops computational systems that can perform some related tasks.

1.2 What are cognition and cognitive science?

Definition 1.1 (Cognition)**.** *Cognition* refers to the processes used to acquire, interpret, store, retrieve, and use information. Examples include perception, attention, memory, learning, language, reasoning, problem solving, and decision making.

Definition 1.2 (Cognitive science)**.** *Cognitive science* is the interdisciplinary scientific study of cognition—how minds and intelligent systems process information. It combines ideas and evidence from psychology, neuroscience, computer science, linguistics, philosophy, and mathematics.

Remember the difference: *Cognition* is what the mind does; *cognitive science* is the field that studies how it works.

Worked Example

Finding a classroom: a step-by-step example.

Process	What the student does
Perception	Reads "Room 204" on the classroom sign.
Attention	Focuses on the room number instead of unrelated posters.
Memory	Remembers that the class is on the second floor.
Reasoning	Concludes that the stairs are a useful route to the second floor.
Decision making	Chooses to take the stairs rather than the lift.
Action and feedback	Walks upstairs; checks whether the room number is correct.

Important: This is an explanatory sequence, not a claim that the brain completes each process in a fixed order. Perception, attention, and memory can interact at the same time.

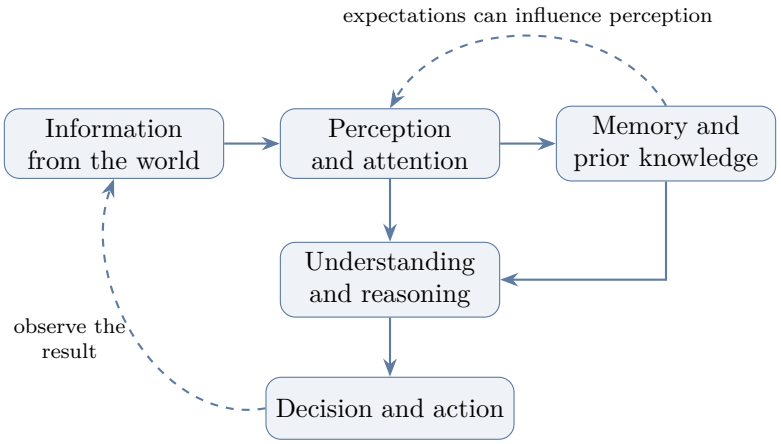


Figure 1.1: A beginner-friendly view of cognition: information, knowledge, and actions influence one another. The arrows show possible interactions, not a single mandatory sequence.

1.3 Major cognitive processes

The following processes are closely connected. We distinguish them to make their roles easier to understand.

1.3.1 Perception

Perception is the interpretation of information from the senses, such as sight or hearing. For example, you see black marks on a screen and recognise them as the words "Room 204." Perception is not simply recording an image: context and expectations can affect interpretation. In AI, computer vision and speech recognition tackle some analogous input-processing tasks.

1.3.2 Attention

Attention helps us prioritise information when many things compete for our focus. While looking for Room 204, you concentrate on room numbers and ignore unrelated advertisements. *Selective attention* focuses on relevant items; *sustained attention* maintains focus over time.

Common Misconception

Human attention and transformer attention are not the same mechanism. In transformer-based AI, "attention" usually refers to a mathematical method for weighting information. It is an analogy to aspects of selection, not a complete model of human concentration.

1.3.3 Memory and learning

Memory enables information to be retained and retrieved. *Learning* is a change in knowledge or behaviour arising from experience. You might *remember* today's room number, then *learn* after several visits that the AI classes usually take place in the same building. A useful beginner distinction is:

- **Sensory memory:** very brief retention of sensory information.
- **Working memory:** temporarily keeping and using a small amount of information, such as "second floor, Room 204."
- **Long-term memory:** knowledge and experiences retained over longer periods, such as the campus layout.

Unlike human learning, a typical machine-learning system learns by adjusting parameters or rules according to training data and an objective.

1.3.4 Language

Language allows us to communicate and interpret structured messages. For example, "The lecture has moved to Room 204" tells you that the location has changed; the meaning depends on the words and their context. Cognitive science examines speech, vocabulary, grammar, meaning, and language use. AI studies related computational problems in natural-language processing.

1.3.5 Reasoning, problem solving, and decision making

These concepts are related but not interchangeable:

- **Reasoning:** drawing a conclusion from information. "Room 204 is on the second floor, so I should go upstairs."
- **Problem solving:** finding steps to achieve a goal. "How do I reach Room 204 from the entrance?"
- **Decision making:** choosing among alternatives. "Should I use the stairs or the lift?"

Human reasoning is developed in Chapter 3; computational representations of reasoning appear later in the course.

Student Activity

Two-minute classroom activity. Your phone rings while you are reading the lecture slides. Name (i) the information competing for attention, (ii) a fact you might retrieve from memory, and (iii) one decision you might make. Compare your answers with a classmate.

1.4 Cognitive science is interdisciplinary

No single discipline completely explains cognition. Different fields ask different questions about the same activity.

Discipline	Example question about finding Room 204
Psychology	How does distraction affect a student's attention?
Neuroscience	What neural mechanisms support memory and navigation?
Computer science	How could a program calculate a route to the classroom?
Linguistics	How does a student understand the message "Go to Room 204"?
Philosophy	What counts as knowing that the room is on the second floor?
Mathematics	How can we represent routes, distances, or uncertainty precisely?

Table 1.1: Six disciplines contribute different tools and questions to cognitive science.

Key Idea

An *observation* describes what someone does (for example, taking the stairs). A *theory or model* proposes how or why that behaviour happens. Researchers test models against evidence, rather than treating a plausible story as proof.

1.5 Three ways to model cognition

A *model* is a simplified representation used to explain or predict some aspect of a system. A model is useful because it focuses on relevant features; it does not need to reproduce every detail of the human brain.

Consider a common computer-science task: **deciding whether an email is spam**. Each modelling approach represents information differently.

1.5.1 Symbolic models: explicit rules

Symbolic models represent knowledge using written facts, logical statements, rules, or graphs. A basic spam checker could contain the rule:

> *IF an email contains "claim your prize" AND has a suspicious link, THEN flag it for review.*

The program applies an explicit rule to the email. This is easy to explain, but new spam messages may avoid those exact words. Symbolic approaches are particularly useful when the rules are well defined.

1.5.2 Connectionist models: learned patterns

Connectionist models use networks of simple numerical units. In machine learning, a neural network can be trained with examples of spam and ordinary emails. Rather than being given the exact rule above, it adjusts many numeric *weights* to detect patterns in its training examples. These models may handle varied wording, but their internal decisions can be harder to explain.

1.5.3 Probabilistic models: uncertain conclusions

Probabilistic models express how strongly evidence supports a conclusion. For example, a model could estimate

$$P(\text{spam} \mid \text{email features}) = 0.80.$$

This means an estimated **80% probability of spam**, given the features used by that model; it does *not* mean that the email is definitely spam. The number is illustrative, not the result of a real experiment. Such models make uncertainty explicit.

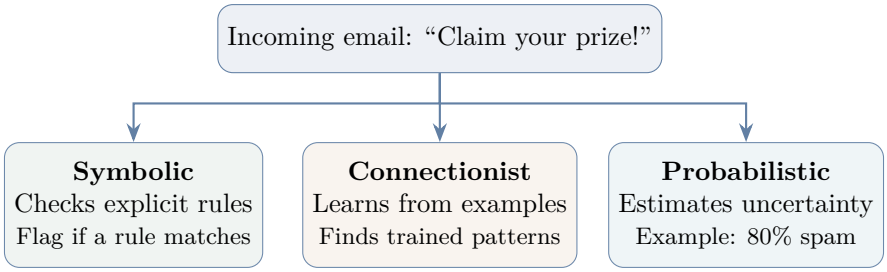


Figure 1.2: One task, three different modelling ideas. The models are illustrative alternatives, not a required three-step pipeline.

	Symbolic	**Connectionist**	**Probabilistic**
Represents	Rules and symbols	Numbers and learned weights	Probabilities
Works by	Applying explicit rules	Learning patterns from examples	Combining evidence with uncertainty
Strength	Often easy to inspect	Can handle complex patterns	Expresses uncertainty
Limitation	Rules may miss new cases	Harder to interpret	Requires suitable probability assumptions or estimates

Table 1.2: A first comparison of three broad modelling traditions.

Common Misconception

These model families are *not mutually exclusive*: a practical AI system might use a neural network to extract features, probabilities to measure uncertainty, and symbolic rules to apply safety requirements.

1.6 What are formalisms, and why do we use them?

A *formalism* is a precise language (often mathematical) for describing a problem. A *cognitive process* is an ability such as remembering; a *model* is a simplified explanation of that ability; and a *formalism* provides notation for expressing the model.

Formalism	**Simple purpose or example**
Logic	Express a rule: “If the building closes, entry is not allowed.”
Probability	Express uncertainty: $P(\text{rain}) = 0.30$ means a 30% chance.
Graphs	Represent rooms as nodes and hallways as connections.
Linear algebra	Store features as numbers, e.g., a vector of measurements.
Calculus	Describe change and how to adjust a numerical model.

Table 1.3: Mathematical tools introduced intuitively; the course develops them later.

Worked Example

Understanding a fire alarm: different kinds of questions.

- A **symbolic safety rule** says: "If the evacuation alarm sounds, follow the building evacuation procedure." This is a policy for action; it does not establish that a fire is certainly present.
- A **probabilistic question** asks: "How likely is there to be a fire, given that the alarm sounded?"
- A **neural classification task** asks: "Can a model recognise an alarm sound from an audio recording?"

The same real-world event motivates different representations and questions.

1.7 How cognitive science connects to artificial intelligence

AI builds computational systems that perform tasks such as recognising patterns, using language, making predictions, planning, and choosing actions. Cognitive science studies these and related abilities in humans and other organisms. The fields influence each other, but their aims are not identical: AI may prioritise useful performance, whereas cognitive science often seeks an explanation of natural cognition.

Connection

Where these ideas lead in this course:

- Human reasoning (Chapter 3): how people draw conclusions.
- Knowledge representation (Chapter 4): explicit facts and rules.
- Neural networks (Chapter 5): learned numerical patterns.
- Reasoning under uncertainty (Chapter 6): probabilistic reasoning.

Artificial neural networks are *loosely inspired* by biological nervous systems; they are not literal copies of the brain.

Student Activity

From human cognition to an AI application. Imagine a campus-navigation app that recommends a route to a lecture room. Discuss: (1) What information does the app need? (2) What can be represented as a graph? (3) What uncertainty might arise? (4) Which aspects of your own navigation—for example, attention or remembering landmarks—are different from what the app computes?

1.8 Chapter recap and practice

Chapter Summary

- **Cognition** is information processing involved in activities such as perceiving, remembering, reasoning, and deciding.
- **Cognitive science** studies cognition using several disciplines.
- Cognitive processes interact; a diagram is a simplified teaching tool.

- **Symbolic models** use rules, **connectionist models** learn numerical patterns, and **probabilistic models** represent uncertainty.
- A **formalism** is a precise way to express a model.
- AI and cognitive science share concepts, but an AI algorithm need not work the same way as a human brain.

Review Questions

Level 1: Recall.

1. In one sentence each, define cognition and cognitive science.
2. Name *at least five* cognitive processes covered in this chapter.
3. Which modelling tradition explicitly represents probabilities?

Level 2: Understand.

4. Explain the difference between memory and learning with one example.
5. Why should Figure 1.1 not be read as a fixed sequence of steps?
6. Explain why an AI attention mechanism is not identical to human attention.

Level 3: Apply.

7. A student uses a bus-tracking app. Identify perception, memory, reasoning, and decision making in the student's behaviour.
8. Describe how a symbolic rule, a learned neural model, and a probability estimate could each help identify spam emails.
9. A navigation app has a map but temporarily loses GPS reception. What information is uncertain? Name one helpful formalism.

Further Reading

For additional reading: Friedenberg & Silverman, *Cognitive Science: An Introduction*; Eysenck & Keane, *Cognitive Psychology: A Student's Handbook*; Russell & Norvig, *Artificial Intelligence: A Modern Approach*, Chapters 1–2. These are enrichment references; students do not need to read them to complete the introductory examples.